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Project

CEH Lab Report – Password Strength Assessment Using Cracklib

Three way to Check Password Strength

- **Cracklib**
- **Python in window using Vscode**
- **Python program in kali linux**

1.Cracklib

Lab Title

Password Strength Assessment Using Cracklib in Kali Linux

Objective

To assess and evaluate password strength using the Cracklib utility in Kali Linux and identify weak passwords that may be vulnerable to password attacks.

Requirements

Hardware Requirements

- Computer or Laptop
- Minimum 4 GB RAM

Software Requirements

- Kali Linux
- Cracklib Runtime Package
- Terminal Access

Commands Used

Update Package Repository:

```
sudo apt update
```

Install Cracklib:

```
sudo apt install cracklib-runtime -y
```

```

(fsociety@Mr)-[~]
$ sudo apt install cracklib-runtime
[sudo] password for fsociety:
The following packages were automatically installed and are no longer required:
  openjdk-21-jre openjdk-21-jre-headless
Use 'sudo apt autoremove' to remove them.

Upgrading:
  cracklib-runtime libcrack2

Summary:
  Upgrading: 2, Installing: 0, Removing: 0, Not Upgrading: 2058
  Download size: 188 kB
  Space needed: 0 B / 29.4 GB available

Continue? [Y/n] y
Get:1 http://http.kali.org/kali kali-rolling/main amd64 libcrack2 amd64 2.9.6-5.2+b2 [44.4 kB]
Get:2 http://http.kali.org/kali kali-rolling/main amd64 cracklib-runtime amd64 2.9.6-5.2+b2 [144 kB]
Fetched 188 kB in 41s (4,622 B/s)
(Reading database ... 430719 files and directories currently installed.)
Preparing to unpack .../libcrack2_2.9.6-5.2+b2_amd64.deb ...
Unpacking libcrack2:amd64 (2.9.6-5.2+b2) over (2.9.6-5.2+b1) ...
Preparing to unpack .../cracklib-runtime_2.9.6-5.2+b2_amd64.deb ...
Unpacking cracklib-runtime (2.9.6-5.2+b2) over (2.9.6-5.2+b1) ...
Setting up libcrack2:amd64 (2.9.6-5.2+b2) ...
Setting up cracklib-runtime (2.9.6-5.2+b2) ...
Processing triggers for doc-base (0.11.2) ...
Processing 2 changed doc-base files ...
Processing triggers for libc-bin (2.42-13) ...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for kali-menu (2025.4.3) ...

```

Start Password Check:

cracklib-check

And then enter Password to check strength

Hello613@

```

(fsociety@Mr)-[~]
$ cracklib-check
Hello613
Hello613: it is based on a dictionary word
Hello7613@
Hello7613@: OK
Pakistan613@
Pakistan613@: OK
PNY_Arfa_Karim
PNY_Arfa_Karim: OK

```

Check a Specific Password:

echo "Password123" | cracklib-check

```
(fsociety@Mr)-[~]  
$ echo "Hello613q" | cracklib-check  
Hello613q: OK
```

Procedure

1. Open the Kali Linux Terminal.
2. Update the package repository.
3. Install the Cracklib Runtime package.
4. Run the `cracklib-check` command.
5. Enter different passwords for testing.
6. Observe the output generated by Cracklib.
7. Record the results and classify passwords as weak or strong.

Observations

Password	Cracklib Result
password123	it is based on a dictionary word
admin123	it is based on a dictionary word
Welcome2026	OK
Abdul@2026	OK
A9#kL2@xP7!mQ4\$z	OK

Results

Cracklib successfully analyzed the passwords and identified weak passwords based on dictionary words, predictable patterns, and insufficient complexity. Strong passwords containing a combination of uppercase letters, lowercase letters, numbers, special characters, and adequate length were accepted.

Conclusion

The password strength assessment was successfully performed using Cracklib in Kali Linux. The tool effectively detected weak passwords and demonstrated the importance of using strong, complex passwords to enhance system security.

Outcome

Learned how to install and use Cracklib in Kali Linux to evaluate password strength and identify weak passwords.

CEH Lab Report – Password Strength Checker Using Python in Visual Studio Code

2. Python in Visual Studio Code

Lab Title

Password Strength Assessment Using Python in Visual Studio Code

Objective

To design and implement a Password Strength Checker using Python in Visual Studio Code and evaluate password security based on length, uppercase letters, lowercase letters, digits, and special characters.

Requirements

Hardware Requirements

- Computer/Laptop
- Minimum 4 GB RAM

Software Requirements

- Windows 10/11
- Visual Studio Code (VS Code)
- Python 3.x
- Python Extension for VS Code

Tools Used

- Visual Studio Code
- Python 3

- Regular Expressions (re Module)

Procedure

1. Install Python 3 on Windows.
2. Install Visual Studio Code.
3. Open VS Code and create a file named password_checker.py.
4. Copy and paste the Python code into the file.
5. Save the file.
6. Open the VS Code Terminal.
7. Execute the program using Python.
8. Enter different passwords and observe the results.

Program Logic

The program checks:

- Password length must be at least 8 characters.
- Password must contain at least one digit.
- Password must contain at least one uppercase letter.
- Password must contain at least one lowercase letter.
- Password should contain at least one special character.

Sample Results

Sample Input and Output

Test Case 1

Input:

Password

```
PS C:\Users\Abdul_Rehman> & C:/Users/Abdul_Rehman/AppData/Local/Programs/Python/Python315/python.exe "d:/# password strength checker.py"
Welcome to the Password Strength Checker!

Enter your password (or type 'exit' to quit): password
Weak: Password must include at least one number.
```

Output:

Weak: Password must include at least one number.

Test Case 2

Input:

Password1

```
Enter your password (or type 'exit' to quit): Password1
Medium: Add special characters to make your password stronger.

Enter your password (or type 'exit' to quit):
```

Output:

Medium: Add special characters to make your password stronger.

Test Case 3

Input:

Password1@

```
Enter your password (or type 'exit' to quit): Password1@
Strong: Your password is secure!

Enter your password (or type 'exit' to quit):
```

Output:

Strong: Your password is secure!

Results

Password	Result
password	Weak
Password1	Medium
Password1@	Strong
Admin123	Medium
Admin123@	Strong

Result

The Password Strength Checker successfully evaluated passwords and classified them according to the implemented security requirements.

Conclusion

The Python application was successfully developed and executed in Visual Studio Code on Windows. The program demonstrated how password complexity can be assessed and how stronger passwords improve security.

Outcome

Successfully created and tested a Password Strength Checker using Python in Visual Studio Code and learned the importance of secure password creation.

CEH Lab Report

3. Python in kali linux

Experiment Name

Password Strength Checker Using Python in Kali Linux Terminal

Aim

To design and implement a Python program in Kali Linux that checks the strength of a password based on security rules.

Objective

To evaluate password strength by verifying:

- Minimum length of 8 characters
- At least one digit
- At least one uppercase letter
- At least one lowercase letter
- At least one special character

Requirements

Hardware Requirements

- Laptop/PC
- Minimum 4 GB RAM

Software Requirements

- Kali Linux
 - Python 3
 - Nano Editor
 - Terminal
-

Program Code

```
# password strength checker

import re

# password strength check conditions:
# min 8 chars, digit, uppercase, lowercase, special char

def check_password_strength(password):
    """
    Function to check the strength of a password.
    """
    if len(password) < 8:
        return "Weak: Password must be at least 8 characters long."

    if not any(char.isdigit() for char in password):
        return "Weak: Password must include at least one number."

    if not any(char.isupper() for char in password):
        return "Weak: Password must include at least one uppercase letter."

    if not any(char.islower() for char in password):
        return "Weak: Password must include at least one lowercase letter."

    if not re.search(r'[@#$%^&*(),.?":{}|<>]', password):
        return "Medium: Add special characters to make your password stronger."

    return "Strong: Your password is secure!"

def password_checker():
    """
    Main function to take user input and check password strength.
    """
    print("Welcome to the Password Strength Checker!")

    while True:
        password = input("\nEnter your password (or type 'exit' to quit): ")

        if password.lower() == "exit":
            print("Thank you for using the Password Strength Checker! Goodbye!")
            break
```

```
        result = check_password_strength(password)
        print(result)

# Run the password checker
if __name__ == "__main__":
    password_checker()
```

Commands Used

Create file:

```
nano password_checker.py
```

Run program:

```
python3 password_checker.py
```

Save & Exit Nano:

- Ctrl + O → Save
 - Enter → Confirm
 - Ctrl + X → Exit
-

Procedure

1. Open Kali Linux Terminal
 2. Create Python file using Nano editor
 3. Paste the program code
 4. Save and exit Nano
 5. Run the program using Python 3
 6. Enter different passwords for testing
 7. Observe results
 8. Exit by typing exit
-

Sample Output

Input:

Password

```
Welcome to the Password Strength Checker!

Enter your password (or type 'exit' to quit): password
Weak: Password must include at least one number.

Enter your password (or type 'exit' to quit): █
```

Output:

Weak: Password must include at least one number.

Input:

Password1

```
Enter your password (or type 'exit' to quit): Password1
Medium: Add special characters to make your password stronger.

Enter your password (or type 'exit' to quit): █
```

Output:

Medium: Add special characters to make your password stronger.

Input:

Password1@

```
Enter your password (or type 'exit' to quit): Password1@
Strong: Your password is secure!

Enter your password (or type 'exit' to quit): █
```

Output:

Strong: Your password is secure!

Result

The Python program successfully evaluates password strength and classifies it as Weak, Medium, or Strong based on defined conditions.

Conclusion

The lab demonstrates how password security can be analyzed using Python in Kali Linux Terminal. Strong passwords require a combination of length, numbers, uppercase, lowercase, and special characters.

Learning Outcome

- Learned Python scripting in Kali Linux
- Understood password security rules
- Practiced conditional logic and loops
- Implemented a real-world security checker